

REFERENCE PRIORS FOR THE MULTIVARIATE CALIBRATION PROBLEM

J.L. du Plessis: Department of Mathematical Statistics, Faculty of
A.J. van der Merwe: Science, University of the Orange Free State, P.O.
and Box 339, Bloemfontein, 9300 Republic of South
P.C.N. Groenewald: Africa

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Summary: Bernardo (1979), proposed the reference prior approach to the development of noninformative priors, the key feature of which was a possible dependence on specification of parameters of interest and nuisance parameters. In this paper the reference prior for the multivariate calibration problem is derived. The theory and results are applied to a real problem.

1. INTRODUCTION

The calibration problem, also called inverse prediction, involves making inferences about a fixed but unknown X corresponding to a response Y . The relationship between Y and X can be expressed as a known linear or non-linear function. Calibration problems have been encountered in a variety of areas within science and engineering and since Krutchkoff's paper appeared in *Technometrics* (1967), a great deal has been written on the problem of calibrating measurement instruments. Univariate Bayesian calibration has been treated by Hoadley (1970) and Hunter and Lamboy (1981). Hoadley's paper was a milestone in the development of statistical calibration theory because it clearly defined a Bayesian approach to univariate calibration and showed that Krutchkoff's inverse regression estimator can be interpreted in a Bayesian way.

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Modern analytical instruments have made it necessary to extend the interest to multivariate calibration. Recently a number of methods for multivariate calibration have been proposed and discussed by Brown (1982), Oman and Wax (1984 and 1985) and Oman (1985) to name but a few.

Determination of reasonable noninformative priors in multiparameter problems is not easy; common noninformative priors, such as the Jeffreys prior, can have an unexpectedly dramatic effect on the posterior. In recognition of this problem, Bernardo (1979), proposed the reference prior approach to the development of noninformative priors, the key feature of which was a possible dependence on specification of parameters of interest and nuisance parameters. This approach was further extended in Berger and Bernardo (1992a, 1992b). In addition, Ghosh and Mukerjee (1992) (see also Clarke, 1989) have shown that the reference priors have optimality properties in terms of asymptotic mean square errors.

In the derivation of the reference priors for the calibration problem similar results and techniques to those given by Kubokawa and Robert (1994) will be used.

2. REFERENCE PRIOR - Σ KNOWN

The calibration experiment can be represented as

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_{1i}, \quad i = 1, \dots, n$$

and the prediction experiment as

$$y_{0j} = \beta_0 + \beta_1 \xi + \varepsilon_{2j}, \quad j = 1, \dots, k.$$

The vectors β_0 , β_1 , y_i and y_{0j} belong to R^p , the x_i 's are known values of the precise measurements and ξ is the value to predict. It is assumed in addition that the error vectors ε_{1i} and ε_{2j} are independent normal variables from $N_p(0, \Sigma)$ and in this section it is also assumed that the covariance matrix Σ is known. Therefore, the aim of the experiment is to predict the quantity ξ based upon $y_{01}, \dots, y_{0k}$ given that β_0 and β_1 have been estimated from the calibration experiment.

The least squares estimate of β_1 is given by $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{c_x}$ where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $c_x = \sum_{i=1}^n (x_i - \bar{x})^2$. Then if $\bar{y}_0 = \frac{1}{k} \sum_{j=1}^k y_{0j}$, the statistics $\hat{\beta}_1$, $\bar{y}$ and $\bar{y}_0$ are minimal sufficient and mutually independent, distributed as

$$(2.1) \quad \hat{\beta}_1 \sim N_p\left(\beta_1, \frac{1}{c_x} \Sigma\right),$$

$$(2.2) \quad \bar{y} \sim N_p\left(\beta_0 + \beta_1 \bar{x}, \frac{1}{n} \Sigma\right),$$

$$(2.3) \quad y_0 \sim N_p\left(\beta_0 + \beta_1 \xi, \frac{1}{k} \Sigma\right).$$

Also

$$(2.4) \quad z = (\bar{y}_0 - \bar{y})n^* \sim N_p\{\beta_1(\xi - \bar{x})n^*, \Sigma\}$$

where

$$n^* = (n^{-1} + k^{-1})^{-\frac{1}{2}}.$$

From the sufficient statistics it follows that the log-likelihood function can be written as

$$(2.5) \quad \ell(\beta_1, \xi) = -\frac{1}{2}\{c_x(\hat{\beta}_1 - \beta_1)' \Sigma^{-1}(\hat{\beta}_1 - \beta_1) + (z - \beta_1(\xi - \bar{x})n^*)' \Sigma^{-1}(z - \beta_1(\xi - \bar{x})n^*)\} + \text{constant}.$$

The reason for writing the log-likelihood function in the form of the sufficient statistics is to facilitate further calculations.

As in the case of the Jeffreys prior, the reference prior method is derived from the Fisher information matrix. To obtain the Fisher information matrix we must calculate the second derivatives of $\ell(\beta_1, \xi)$. Straight-forward calculations lead to the following information matrix for (ξ, β_1) :

$$I(\xi, \beta_1) = \begin{bmatrix} n^{*2} \beta_1' \Sigma^{-1} \beta_1 & n^{*2} (\xi - \bar{x}) \beta_1' \Sigma^{-1} \\ n^{*2} (\xi - \bar{x}) \Sigma^{-1} \beta_1 & \Sigma^{-1} \{c_x + (\xi - \bar{x})^2 n^{*2}\} \end{bmatrix}.$$

By computing the determinant, using the method of partitioned matrices, it follows that

$$|I(\xi, \beta_1)| = \{c_z + (\xi - \bar{x})^2 n^{*2}\}^p |\Sigma|^{-1} \left\{ \frac{n^{*2} \beta_1' \Sigma^{-1} \beta_1 c_z}{c_z + (\xi - \bar{x})^2 n^{*2}} \right\}$$

The Jeffreys prior is then given by

$$\pi^J(\xi, \beta) = |I(\beta_1, \xi)|^{\frac{1}{2}} \propto \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{(p-1)/2} (\beta_1' \Sigma^{-1} \beta_1)^{\frac{1}{2}}$$

The algorithm proposed in Berger and Bernardo (1989) (see also Kubokawa and Robert, 1994) breaks the Fisher information matrix in blocks corresponding to the nuisance parameter β_1 and the parameter of interest, ξ . An explanation of the algorithm will be given in the appendix. It then follows that the reference prior $\pi^R(\xi, \beta_1)$ associated with the ordered group (ξ, β_1) is given by

$$\pi^R(\xi, \beta_1) = \frac{1}{\sqrt{c_z + (\xi - \bar{x})^2 n^{*2}}}.$$

3. THE REFERENCE POSTERIOR DISTRIBUTION OF $\xi - \Sigma$ KNOWN

By completing the square with respect to β_1 in the expression of the likelihood the reference posterior distribution of ξ can be derived as

(3.1)

$$\begin{aligned} p^R(\xi | \text{Data}) &= \int_{\beta_1} L(\beta_1, \xi) \pi^R(\xi, \beta_1) d\beta_1 \\ &\propto (c_z + (\xi - \bar{x})^2 n^{*2})^{-(p+1)/2} \times \\ &\quad \exp \left[\frac{1}{2[c_z + (\xi - \bar{x})^2 n^{*2}]} (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)' \Sigma^{-1} (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z) \right]. \end{aligned}$$

The Jeffreys posterior can be obtained in a similar way.

Since $u = \beta_1' \Sigma^{-1} \beta_1 (c_z + (\xi - \bar{x})^2 n^{*2}) \sim \chi_p^2(\lambda)$ with noncentrality parameter

$$\lambda = \frac{(c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)' (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)}{2[c_z + (\xi - \bar{x})^2 n^{*2}]^2}$$

it follows by integrating with respect to u that the Jeffreys posterior of ξ is given by

(3.2)

$$p^J(\xi | \text{Data}) \propto (c_z + (\xi - \bar{x})^2 n^{*2})^{-1} \exp \left[\frac{1}{2} \frac{(c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)' \Sigma^{-1} (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)}{[c_z + (\xi - \bar{x})^2 n^{*2}]} \right]$$

$$\times \exp \left[-\frac{1}{2} \frac{(c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)' (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)}{[c_z + (\xi - \bar{x})^2 n^{*2}]^2} \right] \times$$

$$\sum_{k=0}^{\infty} \left[\left\{ \frac{(c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)' (c_z \hat{\beta}_1 + (\xi - \bar{x}) n^* z)}{2[c_z + (\xi - \bar{x})^2 n^{*2}]^2} \right\}^k / k! \left[\frac{\Gamma[\frac{1}{2}(p+2k+1)]}{\Gamma[\frac{1}{2}(p+2k)]} \right] \right]$$

3.1 Example

In this example the paint data of Brown (1982) is reanalysed to compare various posterior distributions of ξ . These are derived from the uniform, Jeffreys and reference prior distributions. The posterior distribution from the uniform can easily be obtained by changing the power $-(p+1)/2$ to $-p/2$ in (3.1). As before observations 2, 5, 11, 16, 18, 22, 28, 30 and 35 will be used for prediction purposes and the error covariance matrix will be used as Σ .

Table 3.1 gives the true ξ values for the nine observations as well as the means ($\text{Mean}(\xi)$) and modes ($\text{Mode}(\xi)$) of the three posterior distributions that were used for prediction purposes.

Table 3.1. A comparison of the uniform, Jeffreys and reference posterior densities of ξ

True Value	Uniform		Jeffreys		Reference	
	Mean(ξ_U)	Mode(ξ_U)	Mean(ξ_J)	Mode(ξ_J)	Mean(ξ_R)	Mode(ξ_R)
0	-0.6500	-0.620	-0.7251	-0.680	-0.6400	-0.600
1	0.6997	0.700	0.6876	0.700	0.7020	0.700
2	2.5914	2.558	2.6547	2.610	2.5790	2.540
0	0.4389	0.443	0.4163	0.420	0.4431	0.460
1	0.8646	0.865	0.8578	0.865	0.8656	0.865
2	2.2969	2.278	2.3474	2.313	2.2873	2.260
0	1.2425	1.240	1.2590	1.261	1.2401	1.240
1	1.1053	1.110	1.1108	1.100	1.1045	1.110
2	2.3269	2.300	2.3816	2.348	2.3269	2.280

Inspection of table 3.1 shows that there is not much to choose between the three posterior distributions. The sum of squares loss for the three posterior distributions using the posterior means as predictors are $\Sigma(\xi - \text{Mean}(\xi_U))^2 = 2.8233$, $\Sigma(\xi - \text{Mean}(\xi_J))^2 = 3.1092$ and $\Sigma(\xi - \text{Mean}(\xi_R))^2 = 2.7862$ respectively. In the case of the posterior modes the values are $\Sigma(\xi - \text{Mode}(\xi_U))^2 = 2.7165$, $\Sigma(\xi - \text{Mode}(\xi_J))^2 = 2.9377$ and $\Sigma(\xi - \text{Mode}(\xi_R))^2 = 2.6671$ where U, J and R denote the uniform, Jeffreys and reference priors respectively. For this example $\text{Mode}(\xi)$ is a better predictor of ξ than $\text{Mean}(\xi)$ and the best predictor of ξ is $\text{Mode}(\xi_R)$. In table 3.2 the 95% credibility intervals of ξ are compared.

From table 3.2 it can be seen that the 95% credibility intervals of ξ obtained from the reference posterior density are somewhat shorter than those obtained from the uniform and Jeffreys procedures while containing the true values with the same frequency as the other intervals. In fact, the average lengths of the three intervals are $c_U = 1.5705$, $c_J = 1.5996$ and $c_R = 1.5579$ respectively.

Table 3.2. A comparison of the 95% credibility intervals of ξ for the uniform, Jeffreys and reference posterior distributions

True Value ξ	Uniform 95% credibility interval	Jeffreys 95% credibility interval	Reference 95% credibility interval
0	-1.5142 ; 0.1348	-1.6174 ; 0.0821	-1.4944 ; 0.1440
1	-0.0573 ; 1.4759	-0.0836 ; 1.4441	-0.0542 ; 1.4434
2	1.8280 ; 3.4301	1.8672 ; 3.5192	1.8134 ; 3.4118
0	-0.3166 ; 1.1713	-0.3538 ; 1.1600	-0.3086 ; 1.1725
1	0.1222 ; 1.6280	0.1013 ; 1.6080	0.1259 ; 1.6000
2	1.5568 ; 3.0916	1.5924 ; 3.1636	1.5503 ; 3.0774
0	0.3954 ; 2.1024	0.3936 ; 2.1392	0.3976 ; 2.0954
1	0.3252 ; 1.8905	0.3163 ; 1.9111	0.3274 ; 1.8863
2	1.5800 ; 3.1292	1.6200 ; 3.2050	1.5669 ; 3.1146

4. THE REFERENCE PRIOR OF ξ FOR THE MORE GENERAL CASE $\Sigma = \sigma^2 V$

For this case the sufficient statistics are those given in section 2 together with

$$S = (S_1 + S_2) \sim \sigma^2 \chi_{\frac{q}{2}}^2$$

where

$$S_1 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)' V^{-1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \sim \sigma^2 \chi_{(n-2)p}^2,$$

$$S_2 = \sum_{j=1}^k (y_{0j} - \bar{y}_0)' V^{-1} (y_{0j} - \bar{y}_0) \sim \sigma^2 \chi_{(k-1)p}^2$$

and $\bar{q} = (n-2)p + (k-1)p = (n+k-3)p$.

The log-likelihood is given by

$$l(\xi, \sigma^2, \beta_1) = -(p+q/2) \log \sigma^2 - \left\{ \frac{S + c_z(\beta_1 - \bar{\beta}_1)' V^{-1}(\beta_1 - \bar{\beta}_1) + (z - \beta_1(\xi - \bar{x})n^*)' V^{-1}(z - \beta_1(\xi - \bar{x})n^*)}{2\sigma^2} \right\}$$

up to an additive constant.

Fisher's information matrix for (ξ, β_1, σ^2) ,

$$I(\xi, \sigma^2, \beta_1) = \begin{bmatrix} \frac{n^{*2} \beta_1' V^{-1} \beta_1}{\sigma^2} & 0 & \frac{n^{*2}(\xi - \bar{x})(V^{-1} \beta_1)'}{\sigma^2} \\ 0 & \frac{q+2p}{2\sigma^4} & 0 \dots 0 \\ \frac{n^{*2}(\xi - \bar{x}) V^{-1} \beta_1}{\sigma^2} & 0 & V^{-1} \left[\frac{c_z + (\xi - \bar{x})^2 n^{*2}}{\sigma^2} \right] \\ & \vdots & \\ & 0 & \end{bmatrix}$$

As before, the Jeffreys prior can be obtained by calculating the square root of the determinant of Fisher's information matrix. The result is

$$\pi^J(\xi, \beta_1, \sigma^2) \propto \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{(p-1)/2} (\beta_1' V^{-1} \beta_1)^{\frac{1}{2}} (\sigma^2)^{(p+3)/2}$$

The reference prior $\pi^R(\xi, \sigma^2, \beta_1)$ is given by

$$(4.1) \quad \pi^R(\xi, \sigma^2, \beta_1) = \sigma^{-2} \frac{1}{\sqrt{\{c_z + (\xi - \bar{x})^2\}}}$$

For further details see the appendix.

5. THE REFERENCE PRIOR FOR THE CASE WHERE Σ IS UNKNOWN

To obtain the reference prior for the case where the covariance is unknown, differentiation with respect to Σ or Σ^{-1} must be performed. It is known

from the invariance principal that the two procedures will result in the same Jeffreys posterior. According to Bernardo (personal communication) this will also be the case for the reference posterior. Therefore in this section differentiation will be with respect to the precision matrix Σ^{-1} since it is much easier and as mentioned, the same expression for the posterior will be obtained.

Another statistic must be added to the statistics defined in (2.1)–(2.4) namely that of

$$\mathfrak{Z} = (\mathfrak{Z}_1 + \mathfrak{Z}_2) \sim W(\Sigma, \tilde{q})$$

where

$$\mathfrak{Z}_1 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)' V^{-1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \sim W(\Sigma, n-2),$$

$$\mathfrak{Z}_2 = \sum_{j=1}^k (y_{0j} - \bar{y}_0)' (y_{0j} - \bar{y}_0) \sim W(\Sigma, k-1)$$

and $\tilde{q} = n - 2 + k - 1 = n + k - 3.$

The log-likelihood is then given by

$$\ell(\xi, \Sigma^{-1}, \beta_1) \propto \left(1 + \frac{q}{2}\right) \ell_n \Sigma^{-1}$$

$$- \frac{1}{2} \left\{ c_2 (\beta_1 - \hat{\beta}_1)' \Sigma^{-1} (\beta_1 - \hat{\beta}_1) + (z - \beta_1 (\xi - \bar{x}) n^*)' \Sigma^{-1} (z - \beta_1 (\xi - \bar{x}) n^*) + \text{tr} \Sigma^{-1} \mathfrak{Z} \right\}.$$

The first and second derivatives of $\ell(\xi, \Sigma^{-1}, \beta_1)$ are given in du Plessis et al. (1993).

This leads to Fisher's information matrix for $(\xi, \Sigma^{-1}, \beta_1)$

(5.1)

$$I(\xi, \Sigma^{-1}, \beta_1) = \begin{bmatrix} n^{*2} \beta_1' \Sigma^{-1} \beta_1 & 0 & n^{*2} (\xi - \bar{x}) \beta_1' \Sigma^{-1} \\ 0 & (1 + \frac{1}{2} \tilde{q}) K_p (\Sigma \otimes \Sigma) K_p' & 0 \\ n^{*2} (\xi - \bar{x}) \Sigma^{-1} \beta_1 & 0 & \Sigma^{-1} (c_z + (\xi - \bar{x})^2 n^{*2}) \end{bmatrix}$$

As before, Jeffreys prior can be obtained by calculating the square root of the determinant of Fisher's information matrix. Now

$$|I(\xi, \Sigma^{-1}, \beta_1)| = |I_{22}| |I_{11.2}|$$

where $I_{11} = n^{*2} \beta_1' \Sigma^{-1} \beta_1$, $|I_{22}| = \{c_z + (\xi - \bar{x})^2 n^{*2}\}^p (1 + \frac{1}{2} \tilde{q})^{\frac{1}{2} p(p+1)} |\Sigma|^p$

and $|I_{11.2}| = \beta_1' \Sigma^{-1} \beta_1 \left\{ \frac{n^{*2} c_z}{c_z + (\xi - \bar{x})^2 n^{*2}} \right\}$

Therefore the Jeffreys prior is given by

$$(5.2) \quad \pi^J(\xi, \Sigma^{-1}, \beta_1) = |I_{22}|^{\frac{1}{2}} |I_{11.2}|^{\frac{1}{2}} \alpha \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{\frac{1}{2}(p-1)} |\Sigma|^{\frac{1}{2}p} (\beta_1' \Sigma^{-1} \beta_1)^{\frac{1}{2}}$$

and the reference prior by

$$(5.3) \quad \pi^R(\xi, \Sigma^{-1}, \beta_1) \propto \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{-\frac{1}{2}} |\Sigma|^{\frac{1}{2}(p+1)}$$

The derivations are given in the appendix.

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APPENDIX A

A.1 The reference prior algorithm for the case where Σ is unknown

The reference prior of Berger and Bernardo is motivated by an asymptotic argument, that of maximizing asymptotic missing information (Bernardo, 1979). In the case of a scalar parameter θ the reference prior is the Jeffreys prior, which does have the feature of providing accurate frequentist inference. In the multiparameter setting the situation is much less clear. The reference prior algorithm is relatively complicated and the solution depends on the ordering of the parameters and on how the parameter vector is divided into sub-vectors. In spite of these difficulties, there is growing evidence, mainly through examples, that reference priors provide 'sensible' answers from a Bayesian point of view, and some more limited evidence that frequentist properties of inference from reference posteriors are asymptotically 'reasonable'.

Berger and Bernardo (1989), suggested that one allows multiple groups "ordered" in terms of inferential importance, with the reference prior being determined through a succession of analyses for the implied conditional problems. They particularly recommend the reference prior based on having each parameter in its own group.

In this section the reference prior for the group ordering $\{\xi, \Sigma^{-1}, \beta\}$

will be given. This notation indicates that there are three groups with ξ being the most important.

Let $A(\xi, \Sigma^{-1}, \beta_1) = I^{-1}(\xi, \Sigma^{-1}, \beta_1)$ where $I(\xi, \Sigma^{-1}, \beta_1)$ is the Fisher information matrix as defined in (5.1). Decompose $A(\xi, \Sigma^{-1}, \beta_1)$ as

$$A(\xi, \Sigma^{-1}, \beta_1) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

and define

$$A_i = \begin{bmatrix} A_{11} & A_{1i} \\ \vdots & \vdots \\ A_{i1} & A_{ii} \end{bmatrix}, \quad H_i = A_i^{-1} = \begin{bmatrix} H_{11} & H_{1i} \\ \vdots & \vdots \\ H_{i1} & H_{ii} \end{bmatrix}$$

where $h_i = H_{ii}$.

Now $h_1 = H_{11} = H_1 = A_1^{-1} = A_{11}^{-1}$

where

$$A_{11}^{-1} = n^{*2} \beta_1' \Sigma^{-1} \beta_1$$

$$\begin{aligned} & -[0 n^{*2} (\xi - \bar{x}) \beta_1' \Sigma^{-1}] \begin{bmatrix} (1 + \frac{1}{2} \tilde{q}) K_p' (\Sigma \otimes \Sigma) K_p & 0 \\ 0 & \Sigma^{-1} \{c_x + (\xi - \bar{x})^2 n^{*2}\} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ n^{*2} (\xi - \bar{x}) \Sigma^{-1} \beta_1 \end{bmatrix} \\ & = \beta_1' \Sigma^{-1} \beta_1 \left\{ \frac{n^{*2} c_x}{c_x + (\xi - \bar{x})^2 n^{*2}} \right\} \end{aligned}$$

The reference prior of ξ is

$$\begin{aligned} \pi(\xi) & \propto |h_1|^{\frac{1}{2}} \\ & \propto \{c_x + (\xi - \bar{x})^2 n^{*2}\}^{\frac{1}{2}} \end{aligned}$$

Also

$$A_2 = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \text{ and } H_2 = A_2^{-1} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$$

Therefore

$$H_2 = \begin{bmatrix} n^{*2} \beta_1' \Sigma^{-1} \beta_1 & 0 \\ 0 & (1 + \frac{1}{2} \tilde{q}) K_p' (\Sigma \otimes \Sigma) K_p' \end{bmatrix}$$

$$\begin{bmatrix} n^{*2} (\xi - \bar{x}) \beta_1' \Sigma^{-1} \\ 0 \end{bmatrix} \Sigma \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{-1} [n^{*2} (\xi - \bar{x}) \Sigma^{-1} \beta_1 \ 0]$$

$$h_2 = H_{22} = (1 + \frac{1}{2} \tilde{q}) K_p' (\Sigma \otimes \Sigma) K_p'$$

and

$$\pi(\Sigma^{-1} | \xi) \propto |h_2|^{\frac{1}{2}}$$

$$\propto |\Sigma|^{\frac{1}{2}(p+1)}$$

Further

$$h_3 = H_{33} = \Sigma^{-1} \{c_z + (\xi - \bar{x})^2\} n^{*2}$$

and

$$\pi(\beta_1 | \xi, \Sigma^{-1}) \propto |h_3|^{\frac{1}{2}}$$

$$= \text{constant.}$$

Therefore

$$\pi(\xi, \Sigma^{-1}, \beta_1) = \pi(\xi) \pi(\Sigma^{-1} | \xi) \pi(\beta_1 | \xi, \Sigma^{-1})$$

$$\propto \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{-\frac{1}{2}} |\Sigma|^{\frac{1}{2}(p+1)}$$

as given in equation (5.3).

Since the Jacobian $(\Sigma^{-1} \Rightarrow \Sigma) = |\Sigma|^{-(p+1)}$, the reference prior for

$$\pi(\xi, \Sigma, \beta_1) \propto \{c_z + (\xi - \bar{x})^2 n^{*2}\}^{-\frac{1}{2}} |\Sigma|^{-\frac{1}{2}(p+1)}$$

For $p = 1$ the density corresponds to equation (4.1).